Transformer Models in Natural Language Processing: A Survey

Abstract

This paper surveys transformer architectures including BERT, GPT, and T5. We analyze attention mechanisms, pre-training objectives, and fine-tuning strategies. Our findings show transformers achieve state-of-the-art results across NLP tasks including translation, summarization, and question answering. We identify key challenges including computational cost and data efficiency.

1. Introduction

Transformer architectures have revolutionized NLP since 2017. The self-attention mechanism enables parallel processing of sequences, outperforming RNNs and LSTMs. This survey covers architectural innovations and applications.

2. Methodology

We reviewed 50+ papers on transformer variants. We compared BERT (encoder-only), GPT (decoder-only), and T5 (encoder-decoder) on GLUE, SQuAD, and SuperGLUE benchmarks. Performance metrics include accuracy, F1-score, and inference time.

3. Results

BERT-base achieves 88.5% on GLUE. GPT-3 with 175B parameters shows few-shot learning capabilities. T5 achieves state-of-the-art on 20 of 24 tasks. Training costs remain high at M+ for large models. Efficiency improvements include distillation and quantization.

4. Conclusion

Transformers dominate NLP. Future work includes efficient attention mechanisms, multimodal architectures, and reduced computational requirements.